

ArcticRL: An Autonomous Rover for Ice Sheet Mapping

INTRODUCTION

Climate change is a significant issue, with scientists expecting climate-related deaths projected to rise by **250,000 annually between 2030 and 2050** (NOAA,25). The Arctic and Antarctic regions are the most affected sectors with NOAA predicting a **+7 foot increase** in sea levels in 2100.



Figure 1. Visual of Arctic Ice Sheet (Daily Sabah, 2021)



Figure 2. Remote arctic station (European Space Agency, 2008)

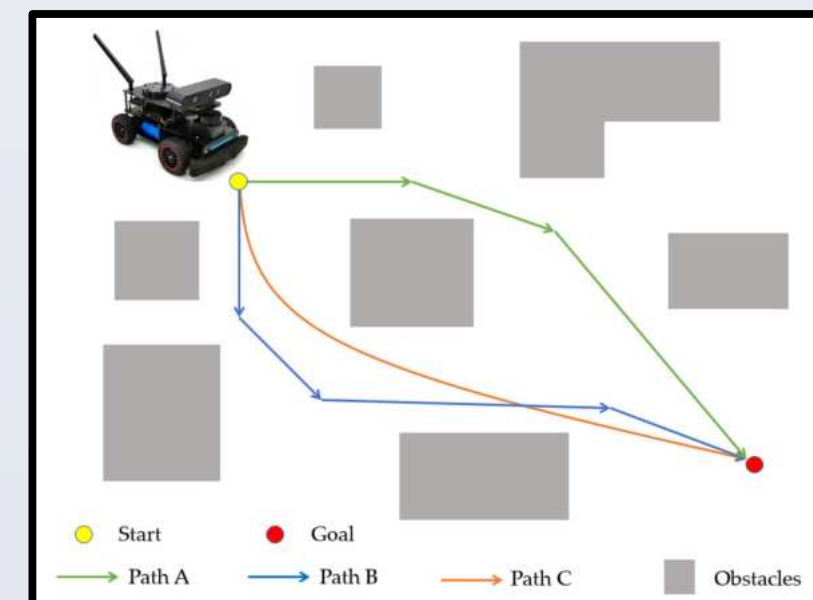


Figure 3. Path Planning in a static environment (Zhang et al., 2022)

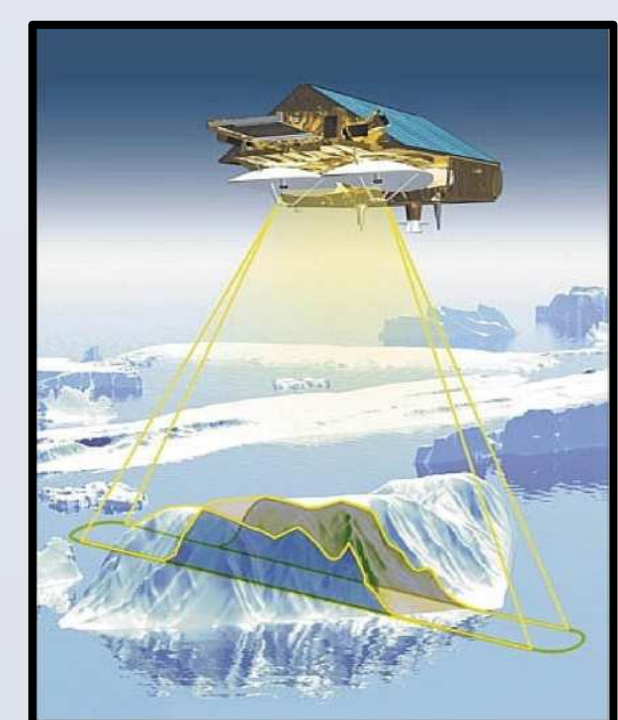


Figure 4. Image of CryoSat-2 (Eoportal, n.d.)

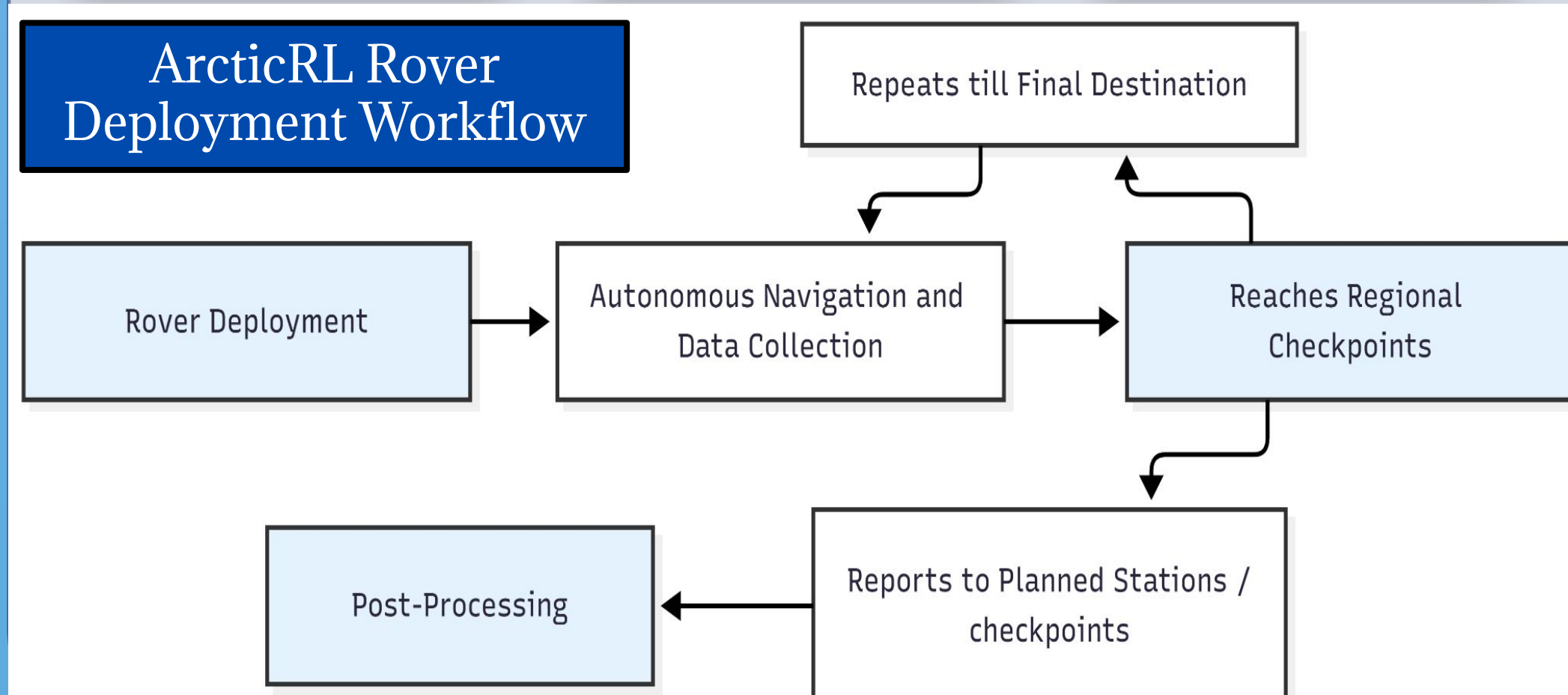
Monitoring ice sheets is crucial as they are massive, sensitive indicators of **global temperature changes**. However, harsh conditions and difficult topographical features prevent researchers from safely and consistently gathering data. This weather and terrain make manual data collection difficult, and limited coverage (**around one station per 70,000 square miles**) also restricts monitoring.

Existing solutions such as satellites like the CryoSat-2, icebreakers like the Polarstern, and autonomous drones like the Vanilla UAV all lack having **both fine level data collection capability and mapping equipment**. Because no current solutions adhere to the constraints and needs of scientists, a new solution must be developed.

SOLUTION OBJECTIVES

RESEARCH QUESTION – How can we create an autonomous rover that is capable of navigating through obstacles and surviving the extreme conditions of the Arctic, while collecting climate data that is crucial for scientific modelling of climate change and global warming?

SOLUTION CRITERIA

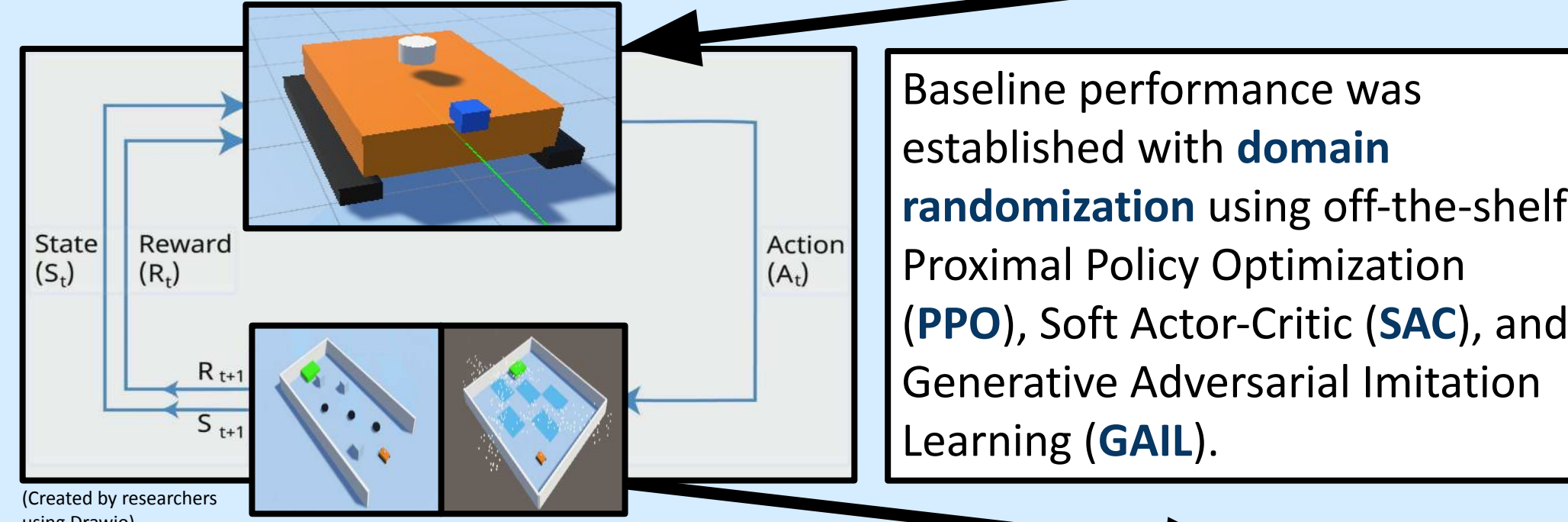
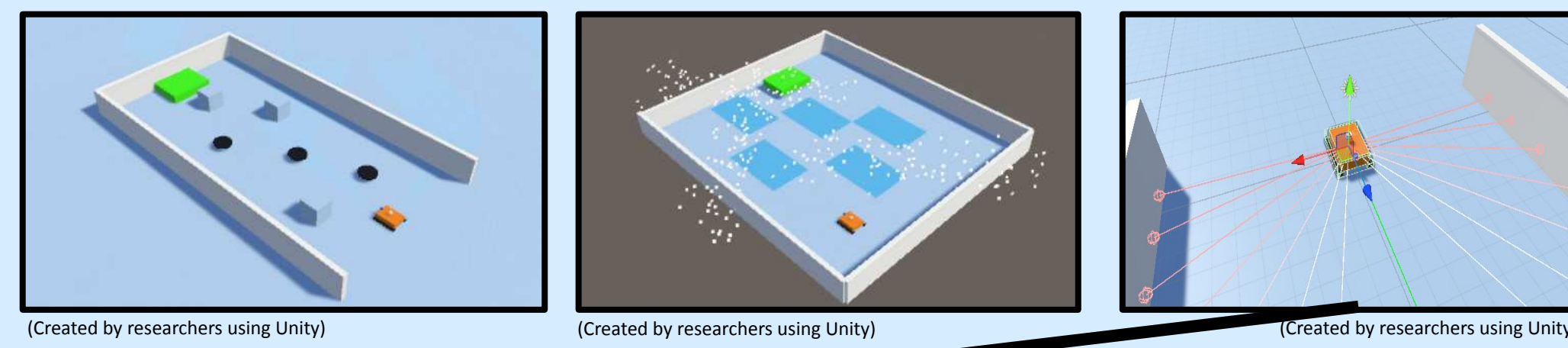


(Created by researchers using Mermaid.io)

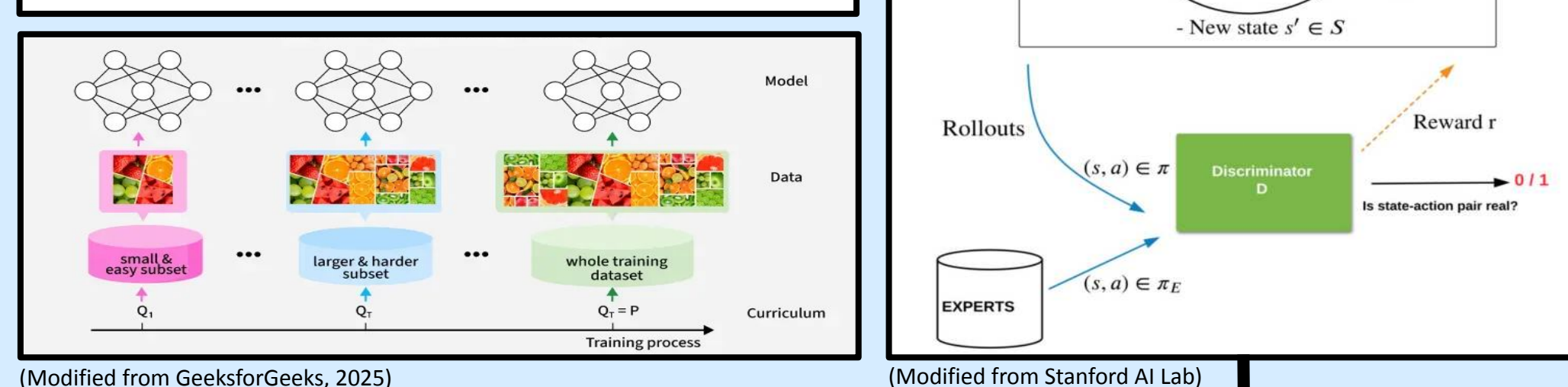
ENGINEERING METHODOLOGY

Part 1: Reinforcement Learning Agent

A **physics-based digital twin training environment** for obstacle avoidance and difficult terrain traversal was created in Unity. The simulated rover agent takes 8 raycast inputs, which map to the real rover's ultrasonic sensors. Domain randomization was added for realistic conditions.



Baseline performance was established with **domain randomization** using off-the-shelf Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Generative Adversarial Imitation Learning (GAIL).



Part 3: Sim2Real

ArcticRL model was exported to a high-performance ONNX format. The physical rover's sensory array was mapped to equivalent raycast sensors to ensure accurate real-time inference with correct state representation.

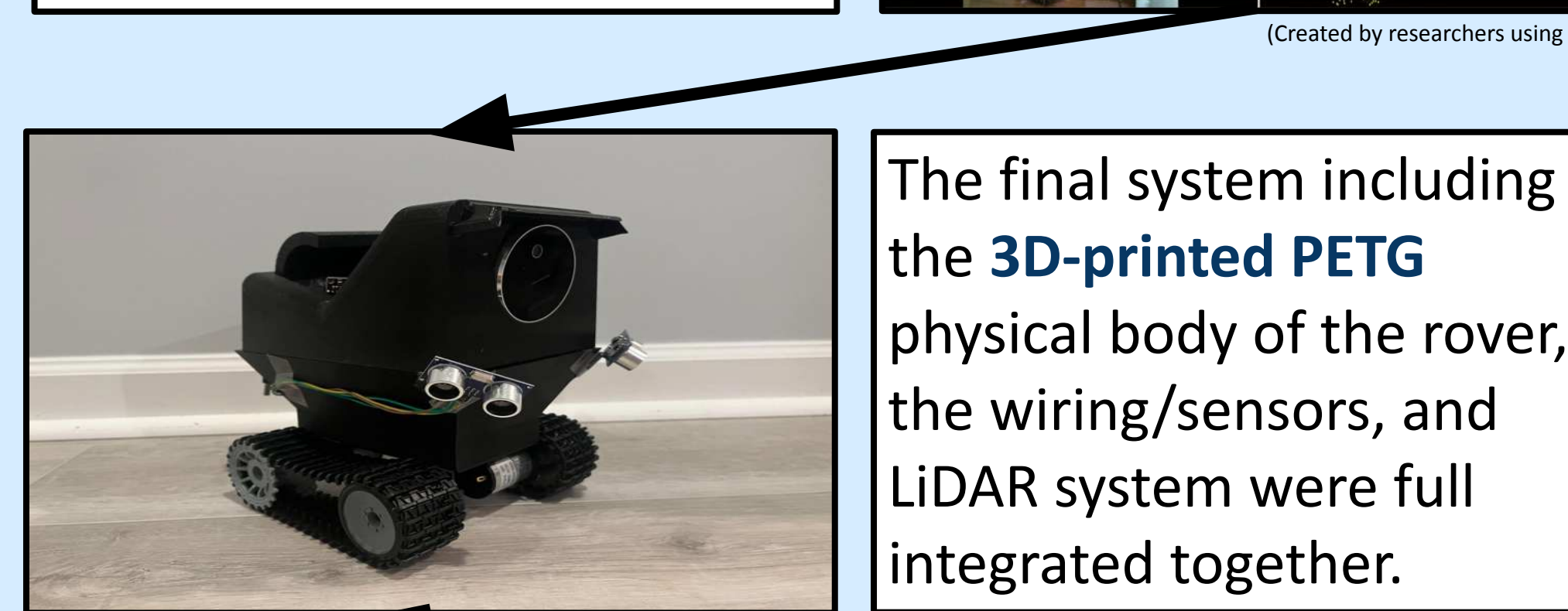
(Modified from EPFL, 2022)

Part 2: Rover Engineering Design

A CAD design was modeled through Inventor, and initial wiring setup for the onboard sensors and **LiDAR** sensor paired through an **Arduino-Raspberry Pi 4** integration was created.



The LiDAR sensor was paired with **RTAB-MAP** and tested real-time on a the Raspberry Pi 4 for real-time 3D terrain mapping.



(Photo taken by authors)

Sensor noise and domain mismatch were mitigated with continuous deployment loops that integrated **real-world telemetry** back into simulation with normalization.

(Modified from Weng, 2019)

RESULTS

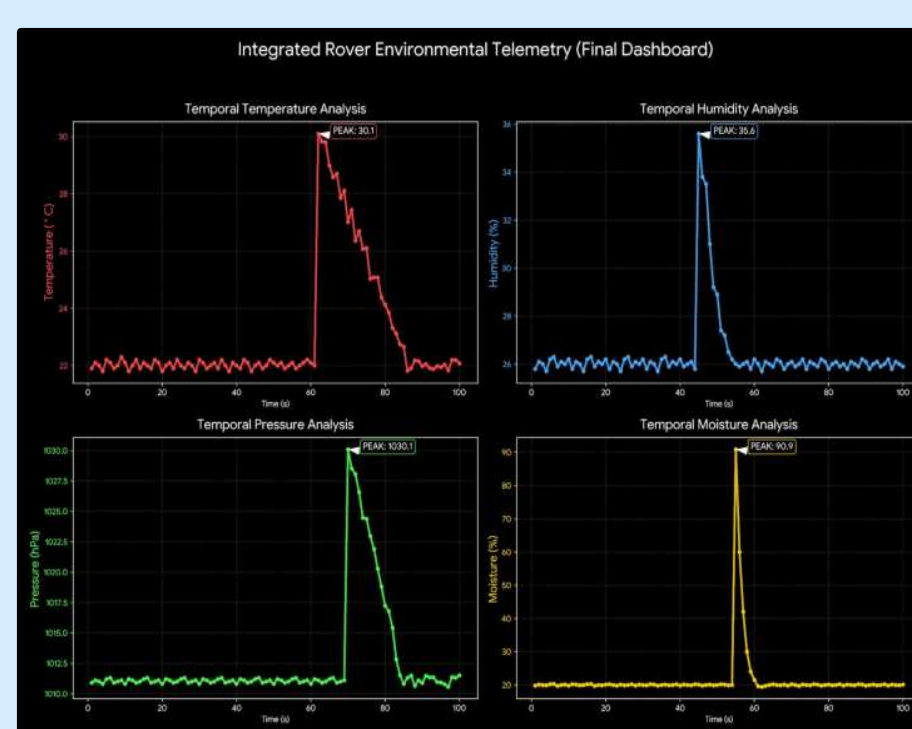


Figure 5. Graphs displaying artificial sensor spikes being recognized

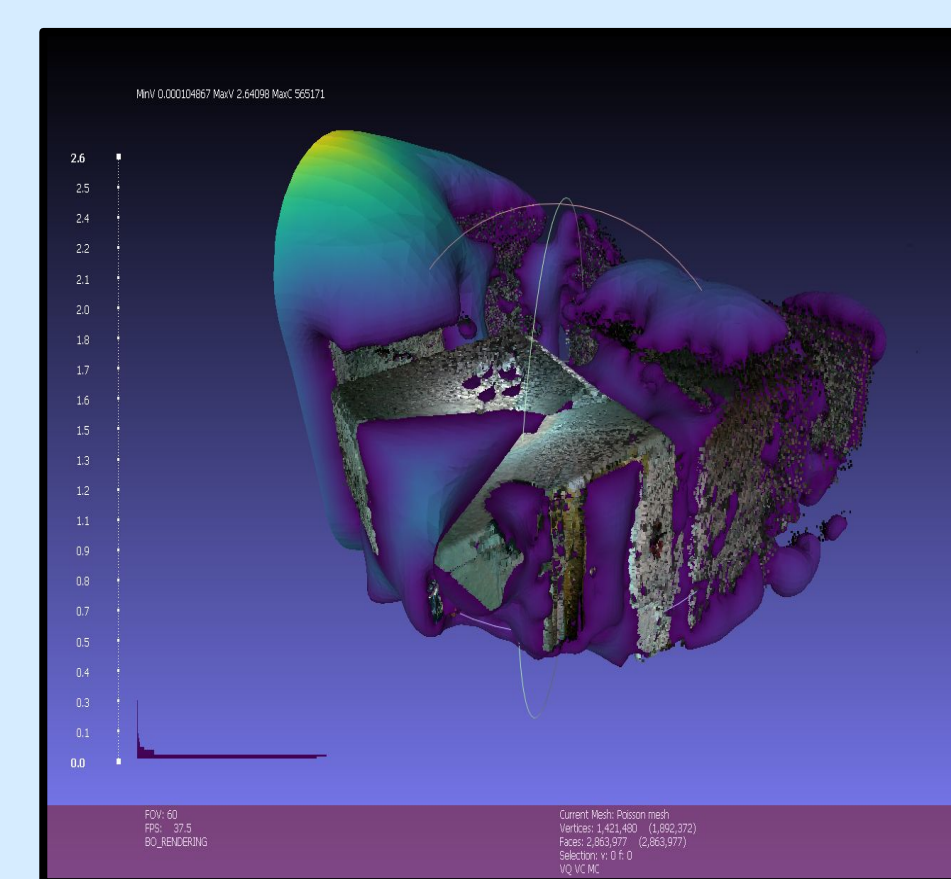


Figure 6. 3D point cloud distance analysis



Figure 7. Point cloud results of a room

A room was mapped and inserted into Meshlab. A comparison of the actual room and the point cloud was created as shown.

Table 1. RL algorithms' performances across various environments across 100 trials

N=100	PPO	SAC	GAIL	ArcticRL (PPO + CL)
Obstacle	0.87	0.85	0.88	0.91
Terrain	0.82	0.81	0.87	0.90
Obstacle (DR)	0.74	0.71	0.62	0.83
Terrain (DR)	0.70	0.68	0.59	0.84

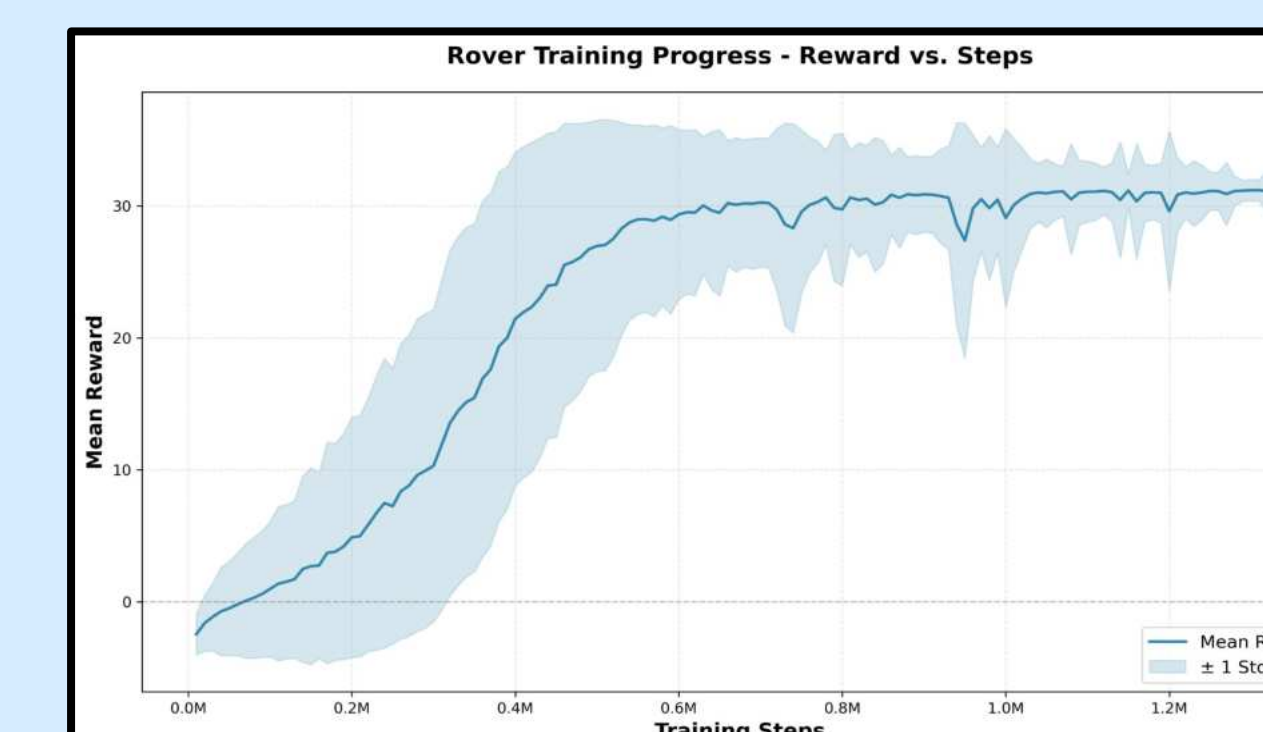


Figure 8. ArcticRL agent training performance

The training progress of ArcticRL was tracked across 1.4M steps, and **mean and std. of reward** was calculated per episode.

DISCUSSION

Physical Rover Data Collection

- The Sensors were able to **accurately identify spikes in data collection** over 5 trials each, with a sample trial being included (Figure 5).
- The Hausdorff distance analysis resulted in only a **maximum error of 0.3**, which is substantially lower compared to other SLAM frameworks.

ArcticRL Model Evaluation

- The training process and average rewards revealed that the model **converged to a high expected reward of ~32** with minimal deviation, showing convergence with an optimal policy
- Integration of **curriculum learning increased success** by up to 14% in random environments.

CONCLUSIONS

- Developed Autonomous navigation system with up to **91% accuracy** in obstacle and complex terrain traversal in Arctic environments
- Succeeded in traversal of randomized environments with accuracies up to **84%**, and successful **Sim2Real transfer** for deployment
- Real-time data collection through onboard sensors and mapping through RTAB-MAP for LiDAR sensor, and accurate point cloud creation

Therefore, ArcticRL's integration within a customized physical rover meets the engineering criteria for deployment in Arctic ice sheets for data collection.

(Retrieved set of icons from Flaticon)

FUTURE WORK

Refine the **Reinforcement Learning agent** and **3D prototype** in future development.

- Develop a more aerodynamic rover to counter harsh winds and Arctic storms
- Further bridge the reality gap between the simulated and deployed rover
- Test in a **real-world environment** (eg. after a snowstorm) and effectively collect data
- Increase the effectiveness of the **sub-glacial lake** detection system
- Develop an interface to visually display collected data

KEY REFERENCES

- Bamber, J. L., Oppenheimer, M., Kopp, R. E., Aspinall, W. P., & Cooke, R. M. (2022). Ice Sheet and Climate Processes Driving the Uncertainty in Projections of Future Sea Level Rise: Findings From a Structured Expert Judgement Approach. *Earth's Future*, 10(10). <https://doi.org/10.1029/2022ef002772>
- Das, S. (2018). *Simultaneous Localization and Mapping (SLAM) using RTAB-MAP*. ArXiv.org. <https://arxiv.org/abs/1809.02989>
- Narvekar, S., Peng, B., Leonetti, M., Sinapov, J., Taylor, M. E., & Stone, P. (2020). Curriculum Learning for Reinforcement Learning Domains: A Framework and Survey. *ArXiv:2003.04960 [Cs, Stat]*. <https://arxiv.org/abs/2003.04960>